

Problem

Design a series RLC circuit with $B = 20$ rad/s and $\omega_0 = 1,000$ rad/s. Find the circuit's Q . Let $R = 10\ \Omega$.

Step-by-step solution

Step 1 of 5

Consider the following data for the series RLC circuit:

Bandwidth of the circuit, $B = 20$ rad/s .

Resonant frequency, $\omega_0 = 1000$ rad/s .

Resistance, $R = 10\ \Omega$.

Step 2 of 5

Write the formula for the bandwidth.

$$B = \frac{R}{L} \dots\dots (1)$$

Here

L Inductor of the series RLC circuit

R resistance of the series RLC circuit

Rewrite the equation (1).

$$L = \frac{R}{B}$$

Substitute 20 for B and 10 for R in the above expression.

$$L = \frac{10}{20} \quad \text{The inductor value for the desired circuit is } \boxed{0.5\text{ H}} .$$

$$= 0.5\text{ H}$$

Step 3 of 5

Write the quality factor expression in terms of bandwidth and resonant frequency.

$$Q = \frac{\omega_0}{B}$$

Substitute 1000 for ω_0 and 20 for B in the above expression

$$Q = \frac{1000}{20}$$

$$= 50$$

Therefore the quality factor of the desired circuit Q is $\boxed{50}$.

Step 4 of 5

Write the quality factor expression in terms of circuit components and resonant frequency.

$$Q = \frac{1}{\omega_0 CR}$$

Rewrite the expression.

$$C = \frac{1}{\omega_0 QR}$$

Substitute 1000 for ω_0 , 10 for R and 50 for Q in the above expression.

$$\begin{aligned} C &= \frac{1}{(1000)(50)(10)} \\ &= 2 \times 10^{-6} \text{ F} \\ &= 2 \mu\text{F} \end{aligned}$$

The capacitor value for the desired circuit is $2 \mu\text{F}$.

Step 5 of 5

Therefore the desired series RLC circuit is shown below.

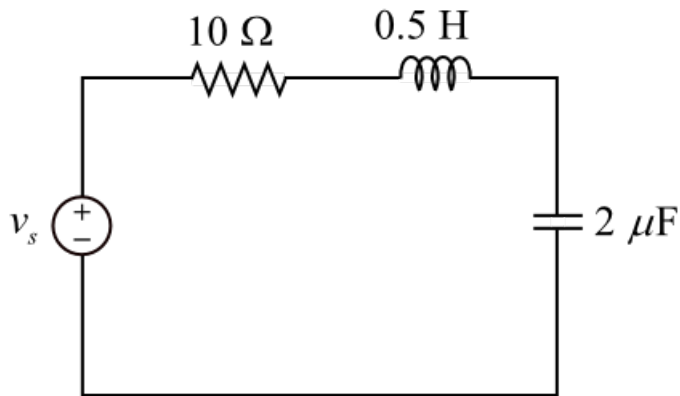


Figure 1